

Shuhan Zheng

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Summary of Qualifications

- Aerospace M.Eng. graduate from the University of Toronto with demonstrated experience in **Space Domain Awareness** (SDA), including the detection and tracking of space objects.
- Practical expertise building **software infrastructure for AI** applications in SDA, published work on **Machine Learning**-enabled SDA techniques.
- Experience **leading** independent and collaborative projects involving orbital dynamics, machine learning, scientific databases, and remote sensing.
- Demonstrated technical communication, project coordination, and stakeholder engagement through conference publications, scientific outreach, student leadership.

Skills

Programming and Analysis: Python, MATLAB, SQL, PyTorch, Pandas, LaTeX, statistical modelling

Aerospace and Simulation Tools: GMAT, SPICE, Orekit, COMSOL Multiphysics, SolidWorks, CFD

Additional Tools: Git, Linux, VS Code, Tableau, Microsoft Office, Power BI

Education

University of Toronto Institute for Aerospace Studies (UTIAS) 2024–2026

M.Eng. in Aerospace Engineering

- Coursework: Spacecraft Dynamics and Control, Control Theory, Computational Fluid Dynamics, Project Management, and Operations Management.

University of Toronto 2020–2024

Honours B.Sc. with double majors in Physics and Astronomy & Astrophysics

Experience

Research Assistant / Supervised Research Project January 2024–August 2026

University of Toronto Institute for Aerospace Studies, Aerospace Mechatronics Lab

- Published technical paper *Dichotomizing Temporarily Bound Objects Using Capture States* on the AMOS 2026 conference as **first author**.
- Implemented and compared machine learning methods, including Probabilistic Neural Network and Multilayer Perceptron, for assisting early classification of asteroids in the Earth-Moon system.

Research Assistant 2023–2024

University of Toronto, Near Field Cosmology Group

- Led development of a SQL-based turnkey solution for automatic retrieval, sanitisation, and cataloguing of stellar data, with a portable interface for machine-learning model ingestion.
- The project was adopted by the Near Field Cosmology Group; the resulting database supports subsequent publications.

Research Assistant Intern Summer 2023

China Spallation Neutron Source, Chinese Academy of Sciences

- Developed a Python-based deterministic simulation tool for generating synthetic polarised-neutron images of magnetised samples.
- Reduced computational time from days to hours compared with conventional Monte Carlo methods.

Space Systems Member 2021–2023

University of Toronto Aerospace Team

- Co-authored FINCH mission design papers for the 2022 and 2024 Small Satellite Conferences.
- Implemented and tested image-correction algorithms for hyperspectral remote-sensing data in collaboration with the optics team for the FINCH microsatellite mission.

Vice-President, Internal and External Affairs

2023–2024

University of Toronto Physics Student Union

- Managed the organisation, logistics, and coordination of outreach, fundraising, social, and academic events; served as a liaison between faculty and students.
- Supported Department of Physics outreach by communicating scientific concepts to students and the public and representing the department at events.

Selected Projects

GFDL - General Flight Dynamics Library

2026

Independent Software Project ([GitHub](#))

- Independently built a modular software platform with swappable backend physics engines and trajectory libraries for deterministic spacecraft mission analysis and flight planning; supports GMAT, SPICE, Orekit, and other tools.
- Interface designed for AI-enabled autonomous orbital flight planning.

Stability and Station-Keeping Near the Earth-Moon L_2 in Planar CRTBP

2025

Nonlinear Control Theory Course Paper

- Modelled spacecraft motion near the Earth-Moon L_2 point using the Circular Restricted Three-Body Problem and assessed stability and station-keeping requirements.
- Implemented Python simulations and designed an LQR controller that stabilised perturbed trajectories, demonstrating closed-loop convergence near an unstable orbital equilibrium.

GMAT Automation for Asteroid Trajectory Prediction

2024

Orbital Dynamics Course Project ([GitHub](#))

- Conducted an individual study on Monte Carlo trajectory prediction of asteroid-Earth encounters.
- Verified trajectory predictions against published journal literature.

Catalogue Acquisition and Compilation Tool

2023

Astronomy Capstone Project ([GitHub](#))

- Independently built an autonomous data-retrieval, sanitisation, and management pipeline with a user-oriented API.
- Designed a scalable scientific database solution for hierarchical and relational data.

Polarisation-Modulated Optical Communication Demonstrator

2023

Electronics Course Project

- Designed and built an optoelectronic prototype for communication over a laser link.

Publications

- Vyse, I., Dagli, R., Chadha, D. V., Ma, J. P., Chen, H., ..., **Zheng, S.**, & Nguyen, K. (2024). *Beyond the Visible: Jointly Attending to Spectral and Spatial Dimensions with HSI-Diffusion for the FINCH Spacecraft*. Small Satellite Conference, Session X: Ground Systems, Utah State University, Logan, UT. <https://digitalcommons.usu.edu/smallsat/2024/all2024/121/>
- Miles, A., Maranto, D., Xu, S., ..., **Zheng, S.**, et al. (2022). *FINCH: A Blueprint for Accessible and Scientifically Valuable Remote Sensing Satellite Missions*. Small Satellite Conference, Weekend Session 7: Science/Mission Payloads – Research & Academia II, Utah State University, Logan, UT. <https://digitalcommons.usu.edu/smallsat/2022/all2022/88/>